1. Answer:

Central Tendency is a statistical measure which is used to descibe the central or average value of a dataset around that gives the characteristics of the entire data which is easier to observe .

The three most common measures of central tendency are:

a. **Mean(average)** : It is calculated by summing up all the values in a dataset and dividing the sum by the total number of values.It is sensitive to extreme values, that makes it susceptible to outliers.

b. **Median:** The median is the middle value in a dataset when the values are arranged in ascending or descending order. If there is an even number of values, the median is the average of the two middle values .It is less affected by outliers as compared to mean.

c. **Mode** : The mode is the value that occurs most frequently in a dataset. It is particularly useful for categorical data but can also be applied to numerical data.In terms of outliers it is less susceptible to outliers as compared to mean but more as compared to Median.

Order of susceptiblity to **outliers** :

Mean > **Mode** > **Median**

The choice of above central tendency measure to use depends on the properties of the data.

For example :

Mean is best for symmetric distributions, while the Median is better for skewed distributions with outliers. The Mode is useful for categorical data.

2 . Answer :

Standard Deviation(SD) is a measure of the amount of variation or dispersion of a set of values from the mean value.

It is calculated as the square root of the variance, which is the average squared deviation from the mean.

A lower value of SD means that most of the numbers are close to the mean (average) value

while a higher value means that the values are spread out over a wider range.

3. Answer :

Box plot is a graphical representation of the distribution of a dataset .

It gives a **five-number** summary of a set of data which is :-

a. **Minimum** – It is the minimum value in the dataset excluding the outliers.

b. **First Quartile (Q1)** – 25% of the data lies below the First (lower) Quartile.

c. **Median (Q2)** – It is the mid-point of the dataset. Half of the values lie below it and half above.It is also known as Second Quartile .

d. **Third Quartile (Q3)** – 75% of the data lies below the Third (Upper) Quartile.

e. **Maximum** – It is the maximum value in the dataset excluding the outliers.

So, by using Box plot we can get a summary of the distribution, identify potential outliers and compare different datasets in a compact and visual manner.

4. Answer :

Interquartile Range(IQR) is used to measure statistical dispersion that represents the middle 50% of a data set which is calculated by dividing a data set into quartiles.

Calculation of IQR :

First we sort dataset in in ascending order and split into 4 equal parts.**Q1**, **Q2**, **Q3** called first, second and third quartiles are the values which separate the 4 equal part.

a. First Quartile (Q1) – 25% of the data lies below the First (lower) Quartile.

b. Second Quartile (Q1) or Median – represents the 50th percentile of the data.

c. Third Quartile (Q3) – represents the 75th percentile of the data.

for e.g.

If a dataset has 2n or 2n+1 data points, then

Q2 = median of the dataset, Q1 = median of n smallest data points and Q3 = median of n

highest data points.

IQR = Q3 – Q1.

The data points which fall below **Q1 – 1.5 IQR** or above **Q3 + 1.5 IQR** are **outliers**.

5. Answer :

Maximum Likelihood Estimation(MLE) is the process of estimating the parameters of a distribution that maximize the likelihood of the observed data belonging to that distribution.

Maximum Likelihood Estimators perform MLE to find the find the distribution that best fits our data and also resulting value of the distribution's parameter is called the **maximum likelihood estimate**.

In this(MLE) after chosing a model for given data distribution , we construct a likelihood function which measures how likely it is to observe our data under the chosen model in first step then we adjust our model parameter to maximize the previous liklihood function.